CHECKING A MORPHISM OF SITES ON A CONSERVATIVE FAMILY OF POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Stalks of a pullback along a point. Same chapter, Lemma lemma-point-pushforward-sheaf. Removes the sheafification in the computation of the stalks of the pullback.
- Stalks as the composite of the point with the functor. Same chapter, Lemma lemma-point-functor. Identifies the stalk of a presheaf with the value at the corresponding functor.
- Conservative families of points. Same chapter, the definition of a conservative family of points. Turns exactness after taking stalks at every member into exactness of the functor.

Source reference. Original equation 04EH: sources/stacks-sites/pinned-original-source.tex, source lines 9433–9433. Navigation pointer invoked at proof line 9433; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00Y8: sources/stacks-sites/pinned-original-source.tex, source lines 9442–9442. Navigation pointer invoked at proof line 9442; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0F4F: sources/stacks-sites/pinned-original-source.tex, source lines 9443–9443. Navigation pointer invoked at proof line 9443; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original definition 00X1: sources/stacks-sites/pinned-original-source.tex, source lines 9448–9448. Navigation pointer invoked at proof line 9448; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $u : \mathcal{C} \to \mathcal{D}$ be a continuous functor of sites. Let $\{(q_i, v_i)\}_{i \in I}$ be a conservative family of points of $\mathcal{D}$. If each functor $u_i = v_i \circ u$ defines a point of $\mathcal{C}$, then u defines a morphism of sites $f : \mathcal{D} \to \mathcal{C}$.*

Proof. Denote p_i the stalk functor (04EH) on $PSh(\mathcal{C})$ corresponding to the functor u_i. We have

$$(f^{-1}\mathcal{F})_{q_i} = (u_s\mathcal{F})_{q_i} = (u_p\mathcal{F})_{q_i} = \mathcal{F}_{p_i}$$

The first equality since $f^{-1} = u_s$, the second equality by Lemma 00Y8, and the third by Lemma 0F4F. Hence if p_i is a point, then pulling back by f and then taking stalks at q_i is an exact functor. Since the family of points $\{q_i\}$ is conservative, this implies that f^{-1} is an exact functor and we see that f is a morphism of sites by Definition 00X1. □